

Electricity

Electric charge and current, potential difference, Ohm's law, resistance and its combination, heating effect of current and electric power, with solved examples.

Electric Charge and Current

Electric charge is a property of matter. There are two types of charge, positive and negative. Like charges repel and unlike charges attract. The SI unit of charge is coulomb, denoted by C. One coulomb is the charge contained in about 6×10^{18} electrons. Charge of one electron is 1.6×10^{-19} C.

Electric current

Electric current is the rate of flow of electric charge. If a charge Q flows through a cross section in time t , then

$$I = Q / t$$

SI unit is ampere, denoted by A. One ampere is one coulomb per second. $1 \text{ A} = 1 \text{ C} / 1 \text{ s}$. Current is measured by an ammeter. An ammeter is always connected in series.

Direction: The direction of conventional current is taken as the direction of flow of positive charge, which is opposite to the direction of flow of electrons. In a circuit, current flows from the positive terminal of the cell to the negative terminal outside the cell.

A continuous and closed path for current to flow is called an electric circuit. A cell or battery provides the energy to move charges.

Potential Difference

Charges do not flow on their own. They need an electric pressure to move. This pressure is called potential difference. It is created by a cell or battery.

Potential difference

Potential difference between two points is the work done to move a unit charge from one point to the other.

$$V = W / Q$$

where V is potential difference, W is work done and Q is charge moved. SI unit is volt, denoted by V. 1 volt = 1 joule / 1 coulomb. It is measured by a voltmeter. A voltmeter is always connected in parallel.

One volt means that one joule of work is done to move one coulomb of charge between two points.

Electric potential

Electric potential at a point is the work done to bring a unit positive charge from infinity to that point. Potential difference is the difference in potential between two points.

Ohm's Law

George Simon Ohm studied the relation between current and potential difference. He found a simple law that holds for many materials, especially metals, when temperature remains the same.

Ohm's Law

At constant temperature, the current through a conductor is directly proportional to the potential difference across its ends.

$$V \text{ is proportional to } I \quad \text{or} \quad V = I R$$

Here R is a constant called resistance of the conductor. SI unit of resistance is ohm, denoted by Greek letter omega. 1 ohm = 1 volt / 1 ampere.

Graph: If we plot V on the y axis and I on the x axis, the graph is a straight line passing through the origin. The slope of this line is R. Conductors that obey Ohm's law are called ohmic conductors. Their V-I graph is a straight line.

Conditions: Ohm's law holds only if temperature stays constant. If temperature changes, resistance changes and the graph is no longer a straight line. Components like diodes and transistors do not obey Ohm's law.

Worked Example

A current of 0.5 A flows through a wire when a potential difference of 3 V is applied. What is the resistance? Also find the current when the potential difference is 9 V.

$$R = V / I = 3 / 0.5 = 6 \text{ ohm}$$

$$\text{When } V = 9 \text{ V, } I = V / R = 9 / 6 = 1.5 \text{ A}$$

Resistance and Resistivity

Resistance is the opposition offered by a conductor to the flow of current. It arises because moving electrons collide with atoms inside the conductor.

Factors on which resistance depends:

- **Length:** Resistance is directly proportional to length. R is proportional to l. A longer wire has more resistance.
- **Area of cross section:** Resistance is inversely proportional to area. R is proportional to $1 / A$. A thicker wire has less resistance.
- **Material:** Different materials have different resistance even if length and area are the same.
- **Temperature:** For metals, resistance increases with temperature.

Combining the first three factors, for a given material at a given temperature:

$$R = \rho \times l / A$$

Here ρ , written as the Greek letter rho, is called resistivity or specific resistance. It depends only on the material and temperature, not on length or area.

Resistivity

Resistivity is the resistance of a conductor of unit length and unit cross sectional area. SI unit is ohm metre, ohm m.

$$\rho = R \times A / l$$

Conductors like copper and aluminium have low resistivity, so they are used for wiring. Alloys like manganin and constantan have higher resistivity and their resistivity changes very little with temperature, so they are used for standard resistors. Alloys like nichrome have high resistivity and high melting point, so they are used for heating elements.

Resistivity values: Silver is the best conductor with lowest resistivity. Glass and rubber have very high resistivity and act as insulators.

Worked Example

A wire of length 2 m and cross section 1 mm^2 has a resistance of 0.5 ohm. Find its resistivity.

$$A = 1 \text{ mm}^2 = 1 \times 10^{-6} \text{ m}^2, \quad l = 2 \text{ m}, \quad R = 0.5 \text{ ohm}$$

$$\rho = R A / l = 0.5 \times 1 \times 10^{-6} / 2 = 0.25 \times 10^{-6} \text{ ohm m}$$

Combination of Resistances

Resistors can be connected in two main ways, in series and in parallel. Many circuits use a mix of both.

Resistors in Series

In series, resistors are connected end to end, so the same current flows through each resistor. Only one path exists for current.

$$R_s = R_1 + R_2 + R_3 + \dots$$

The equivalent resistance is the sum of individual resistances. So it is larger than any single resistance.

Current is the same through each resistor. Potential difference divides across resistors in proportion to resistance: $V = V_1 + V_2 + V_3$.

Worked Example

Two resistors of 3 ohm and 6 ohm are connected in series. Find the equivalent resistance. If a 9 V battery is connected, find the current and the voltage across each resistor.

$$R_s = 3 + 6 = 9 \text{ ohm}, \quad I = V / R_s = 9 / 9 = 1 \text{ A}$$

$$V_1 = I R_1 = 1 \times 3 = 3 \text{ V}, \quad V_2 = 1 \times 6 = 6 \text{ V}, \quad \text{check: } 3 + 6 = 9 \text{ V}$$

Resistors in Parallel

In parallel, one end of all resistors is connected to one point and the other end to another point. So the potential difference is the same across each resistor, but current divides.

$$1 / R_p = 1 / R_1 + 1 / R_2 + 1 / R_3 + \dots$$

For two resistors, $R_p = R_1 R_2 / (R_1 + R_2)$.

The equivalent resistance in parallel is smaller than the smallest individual resistance.

Voltage is the same across each resistor. Total current is the sum of currents through each branch: $I = I_1 + I_2 + I_3$.

Worked Example

Two resistors of 3 ohm and 6 ohm are connected in parallel. Find the equivalent resistance and the total current from a 9 V battery.

$$1 / R_p = 1/3 + 1/6 = 2/6 + 1/6 = 3/6 = 1/2 \quad \text{so} \quad R_p = 2 \text{ ohm}$$

$$I = V / R_p = 9 / 2 = 4.5 \text{ A}$$

Also, $I_1 = 9/3 = 3 \text{ A}$, $I_2 = 9/6 = 1.5 \text{ A}$, total = 4.5 A.

Use: Household wiring is done in parallel so that each appliance gets full voltage and can be switched on and off independently, and the equivalent resistance stays low.

Heating Effect of Current

When current flows through a resistor, electrical energy is converted into heat energy. This is called the heating effect of current or Joule heating. The battery does work to move charges against resistance, and this work appears as heat.

If a current I flows through a resistor R for time t under potential difference V , then:

- Charge moved, $Q = I t$
- Work done, $W = V Q = V I t$
- Since $V = I R$, we can write W or heat produced H as

$$H = V I t = I^2 R t = V^2 t / R \quad (\text{in joule})$$

This is Joule's law of heating. Heat produced is directly proportional to square of current, to resistance, and to time.

Applications: Electric heater, electric iron, electric bulb filament, electric fuse. The filament of a bulb is made of tungsten because it has high resistivity and high melting point. The fuse wire melts when current exceeds a safe value and breaks the circuit, protecting appliances.

Worked Example

A heater of resistance 50 ohm is connected to 220 V for 10 seconds. Find the heat produced.

$$H = V^2 t / R = 220 \times 220 \times 10 / 50 = 9680 \text{ J}$$

$$\text{Or, } I = V/R = 4.4 \text{ A, } H = I^2 R t = 4.4 \times 4.4 \times 50 \times 10 = 9680 \text{ J.}$$

Electric Power and Energy

Electric power

Electric power is the rate at which electrical energy is consumed or produced.

$$P = W / t = V I = I^2 R = V^2 / R$$

SI unit is watt, denoted by W. 1 watt = 1 joule / 1 second = 1 volt x 1 ampere. Larger units are kilowatt, 1 kW = 1000 W, and megawatt.

Power rating on an appliance: A bulb rated 220 V, 100 W means it consumes 100 J per second when operated at 220 V. If voltage is lower, power consumed is lower.

Commercial unit of energy

The commercial unit of electrical energy is kilowatt hour, kWh. One kilowatt hour is the energy consumed when 1000 watt is used for one hour. It is called one unit.

$$1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s} = 3.6 \times 10^6 \text{ J}$$

Energy in kWh = Power in kW x Time in hours. Electricity bill is calculated in units.

Worked Example

A 100 W bulb is used for 5 hours daily. Find the energy consumed in 30 days in kWh and in joule. Cost is Rs 6 per unit. Find the bill.

$$\text{Energy per day} = 0.1 \text{ kW} \times 5 \text{ h} = 0.5 \text{ kWh}$$

$$\text{Energy in 30 days} = 0.5 \times 30 = 15 \text{ kWh} = 15 \text{ units}$$

$$\text{In joule} = 15 \times 3.6 \times 10^6 = 5.4 \times 10^7 \text{ J}, \quad \text{Bill} = 15 \times 6 = \text{Rs } 90$$

Summary

- Charge Q , current $I = Q / t$, unit ampere. Potential difference $V = W / Q$, unit volt. Ammeter in series, voltmeter in parallel.
- Ohm's law: $V = I R$ at constant temperature. V - I graph is a straight line through origin.
- Resistance depends on length, area, material and temperature. $R = \rho l / A$. Resistivity ρ is property of material, unit ohm m.
- Series: $R_s = R_1 + R_2 + \dots$, same current, voltage divides. Parallel: $1 / R_p = 1 / R_1 + 1 / R_2 + \dots$, same voltage, current divides. Parallel gives smaller equivalent resistance.
- Joule heating: $H = I^2 R t = V I t = V^2 t / R$. Used in heaters, bulbs and fuses.
- Power $P = V I = I^2 R = V^2 / R$, unit watt. Energy = $P \times t$. 1 kWh = 3.6×10^6 J, one unit for billing.

Practice Questions

1. Define current and potential difference. State their SI units and the instruments to measure them and how they are connected.

2. State Ohm's law and draw the V-I graph. A wire has resistance 6 ohm. What current flows when 12 V is applied?

3. A wire of length l and cross section A has resistance R . What will be the resistance of a wire of same material with length $2l$ and cross section $A/2$?

4. Three resistors 2 ohm, 3 ohm and 6 ohm are connected (a) in series (b) in parallel. Find the equivalent resistance in each case.

5. An electric bulb is rated 220 V, 60 W. Find its resistance and the current drawn when operated at 220 V. What is the power if operated at 110 V?

6. A heater rated 1000 W, 220 V is used for 2 hours daily for 30 days. How many units of energy are consumed? Find the heat produced in joule.